

i3 SmartLabs — Engineering Runbook & Technical Specs

VPC Peering, KubeVirt Virtual Appliance Manifests, Ansible Automation & SSO Client Specs

Companion Technical Guide to Master Commercialization Blueprint • i3 Technologies Limited • ROKS 4.17

1. CROSS-ACCOUNT IBM CLOUD VPC PEERING & TRANSIT ROUTING

To unify our 3 IBM Cloud accounts (\$3,000/quarter each) into a single logical data plane, we deploy an IBM Cloud Transit Gateway interconnecting Account 1 (ROKS Production Core), Account 2 (ROKS SmartLabs Engine), and Account 3 (Model Gateway & GPU Sandbox).

```
# terraform/modules/transit_gateway/main.tf resource "ibm_tg_gateway" "i3_smartlabs_tgw" { name = "i3-smartlabs-transit-gateway" location = "eu-de" global = false resource_group = var.resource_group_id } # Cross-Account Connection: Account 1 (Control Plane VPC) resource "ibm_tg_connection" "conn_account_1" { gateway = ibm_tg_gateway.i3_smartlabs_tgw.id network_type = "vpc" name = "conn-account1-prod-core" network_id = var.vpc_account_1_crn } # Cross-Account Connection: Account 2 (Compute Engine VPC) resource "ibm_tg_connection" "conn_account_2" { gateway = ibm_tg_gateway.i3_smartlabs_tgw.id network_type = "vpc" name = "conn-account2-compute" network_account_id = var.account_2_id network_id = var.vpc_account_2_crn } # Cross-Account Connection: Account 3 (Model Gateway & GPU VPC) resource "ibm_tg_connection" "conn_account_3" { gateway = ibm_tg_gateway.i3_smartlabs_tgw.id network_type = "vpc" name = "conn-account3-ai-models" network_account_id = var.account_3_id network_id = var.vpc_account_3_crn }
```

2. OPENSIFT VIRTUALIZATION (KUBEVIRT) VM APPLIANCE SPECIFICATION

For vendor appliances requiring full hardware emulation (Fortinet FortiGate, Palo Alto VM-Series, Cisco IOSv), OpenShift Virtualization runs containerized virtual machines within student namespaces alongside container pods:

```
# manifests/kubevirt/fortigate-vm.yaml apiVersion: kubevirt.io/v1 kind: VirtualMachine metadata: name: fortigate-firewall namespace: "lab-{{ student_id }}" spec: running: true template: metadata: labels: kubevirt.io/domain: fortigate-vm spec: domain: cpu: cores: 2 resources: requests: memory: 4Gi devices: disks: - name: boot-disk disk: bus: virtio interfaces: - name: management-net masquerade: {} - name: lab-internal-lan bridge: {} networks: - name: management-net pod: {} - name: lab-internal-lan multus: networkName: student-isolated-br volumes: - name: boot-disk containerDisk: image: image-registry.openshift-image-registry.svc:5000/labs-security/fortigate:7.4.3
```

3. AUTOMATED HEALTH CHECK & TOKEN ACCOUNTING DAEMON

To maintain billing fidelity across corporate tenants, a lightweight cron daemon monitors pod status and reconciles token consumption between LiteLLM and EvalOS databases:

```
#!/usr/bin/env bash # scripts/reconcile_tenant_tokens.sh set -euo pipefail LITELLM_URL="http://litellm-proxy.i3-model-gateway.svc.cluster.local:4000" MASTER_KEY=$(oc get secret litellm-secrets -n i3-model-gateway -o jsonpath='{.data.LITELLM_MASTER_KEY}' | base64 -d) # Fetch tenant token usage report curl -s -X GET "$LITELLM_URL/pend/calculate" -H "Authorization: Bearer ${MASTER_KEY}" -H "Content-Type: application/json" | jq -r '.[ ] | [.user, .spend, .total_tokens] | @csv' > /tmp/daily_token_spend.csv # Ingest into Postgres billing ledger PGPASSWORD=$(oc get secret evalos-web-secrets -n i3-evalos -o jsonpath='{.data.DATABASE_URL}' | base64 -d) psql "$PGPASSWORD" -c "\copy tenant_usage_ledger(tenant_id, spend_usd, total_tokens) FROM '/tmp/daily_token_spend.csv' WITH CSV;" echo "Token accounting synced successfully at $(date -u)"
```

4. OPERATIONAL PRODUCTION SIGN-OFF CHECKLIST

MILESTONE	VERIFICATION COMMAND / CHECK	RESPONSIBLE LEAD	STATUS
Transit Gateway Active	ibmcloud tg gateways shows status available for all 3 VPC connections	Infrastructure Architect	Ready for Q4
Keycloak SSO Federation	Token validation passes across evalos-spa, jupyterhub, and sage	Platform Engineer	Active
LiteLLM Model Fallback	Requests gracefully fall back from external APIs to on-cluster qwen-fast	DevOps Lead	Validated

MILESTONE	VERIFICATION COMMAND / CHECK	RESPONSIBLE LEAD	STATUS
Daraja M-Pesa Callback	STK push callbacks verify through AfroERP webhook endpoints	FinTech Integration Lead	Configured
Ansible Teardown Cron	Namespaces older than 4 hours automatically culled without leaving orphan PVCs	Automation Lead	Production-Ready

Master Architecture & Runbook Signed Off • i3 Technologies Limited • Executive Management Committee